

Software Engineering Laboratory

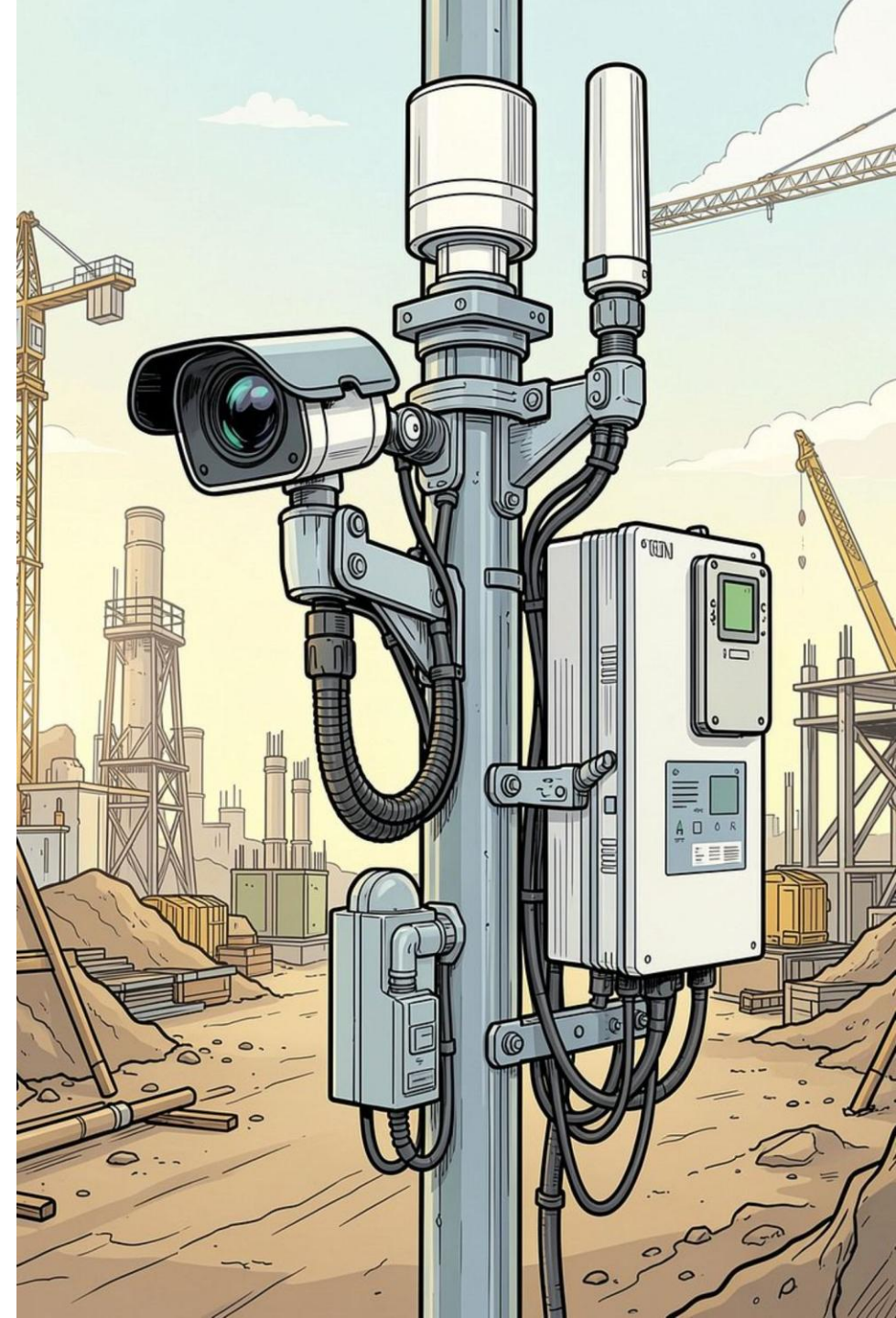
AI-Enabled IoT-Based Real-Time Environmental and Surrounding Safety Monitoring System

SmartSurround

Powered by TEAM ABISKAR

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Intelligent Monitoring · Detection · Prediction · Safety



Problem :-

- **Fragmented Monitoring** – Different parameters require separate systems.
- **Manual Monitoring** – Depends on human observation.
- **Delayed Detection** – Hazards may not be detected early.
- **No Visual Intelligence** – Cannot detect waterlogging or road damage.
- **Manual Reporting** – Incidents are reported manually.
- **Poor Location Tracking** – Accurate GPS information may be unavailable.
- **Reactive Safety** – Action is taken after hazards occur.
- **Limited Data Analysis** – Historical trends are not fully utilized.
- **No Centralized Dashboard** – Multiple systems must be monitored separately.
- **Limited Access Control** – Different user roles may not be properly managed.

Simple Comparison

CURRENT: Manual Monitoring →
Delayed Detection → Manual
Reporting → Reactive Action

NEED: Continuous Monitoring →
Intelligent Detection → Alert →
Early Action

Background

Workers in mines, industry, construction, roads and remote areas face dust, heat, gases, noise, fire, rain and infrastructure hazards.

Current Method

Separate instruments, manual observation, independent systems, and manual reporting
→ limited use of historical data for prediction.

Limitations

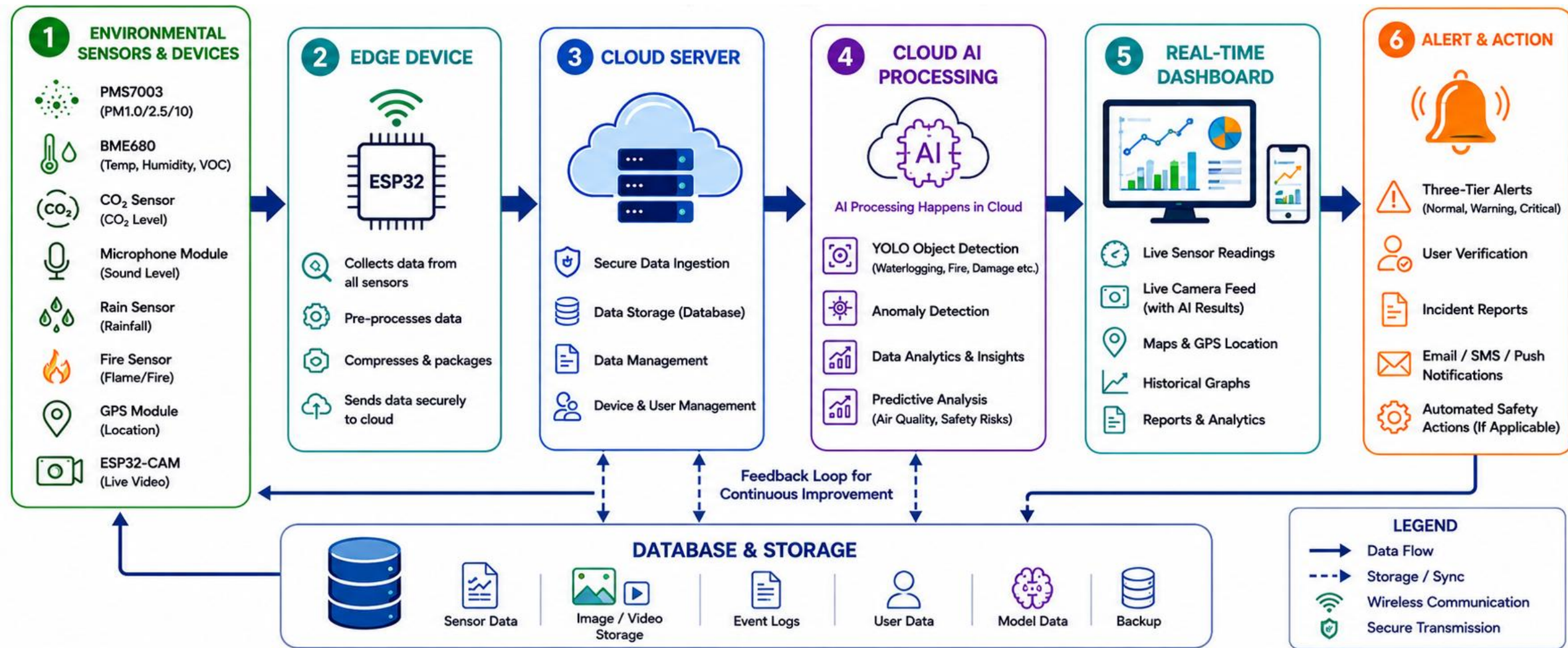
Lack of continuous central monitoring, delayed detection, limited visual intelligence, no GPS-based reporting, weak predictive capability.

Who is Affected

Mine/industrial/construction workers, road users, municipal authorities, people in hazardous or remote areas.

System Architecture & Working Flow

Sensors collect data, the ESP32 transmits it wirelessly, and a web dashboard displays real-time readings with AI analysis triggering alerts and incident reports.



The ESP32-CAM also enables AI-powered visual anomaly detection — identifying fire, waterlogging, damaged roads, and other hazards from live camera imagery.

Technical Feasibility, Cost & Technology Study

Technical Feasibility

- **ESP32 + ESP32-CAM:** IoT control, wireless communication & live video.
- **Sensors:** Monitor air quality, temperature, humidity, VOC, CO₂, sound, rain & fire.
- **GPS:** Provides incident location.
- **Web Dashboard:** Shows real-time data, trends & camera feed.
- **AI:** Detects environmental and safety hazards.

Overall Feasibility: HIGH – Practical and suitable for prototype development.

Cost Study

Component	Cost
ESP32	₹782
ESP32-CAM	₹900
PMS7003	₹1,528
BME680	₹819
Microphone Module	₹100
Rain Sensor	₹200
Fire Sensor	₹50
GPS Module	₹550
Total	₹4,929

Technology Study

Technology	Purpose
ESP32	IoT controller and wireless communication
ESP32-CAM	Live video monitoring
PMS7003	Air-quality/particulate monitoring
BME680	Temperature, humidity and VOC monitoring
CO ₂ Sensor	CO ₂ measurement
Microphone	Sound-level monitoring
Rain Sensor	Rainfall detection
Fire Sensor	Fire/flame detection
GPS	Location and incident reporting
IoT Communication	Wireless sensor-data transmission
Web Dashboard	Real-time monitoring and historical graphs
AI Vision Analytics	Fire, smoke, waterlogging and road-damage detection
AI/ML Analysis	Predictive safety analysis
Alert System	Normal → Warning → Critical alerts

Research Gap and Existing System Comparison

Research Gap Findings

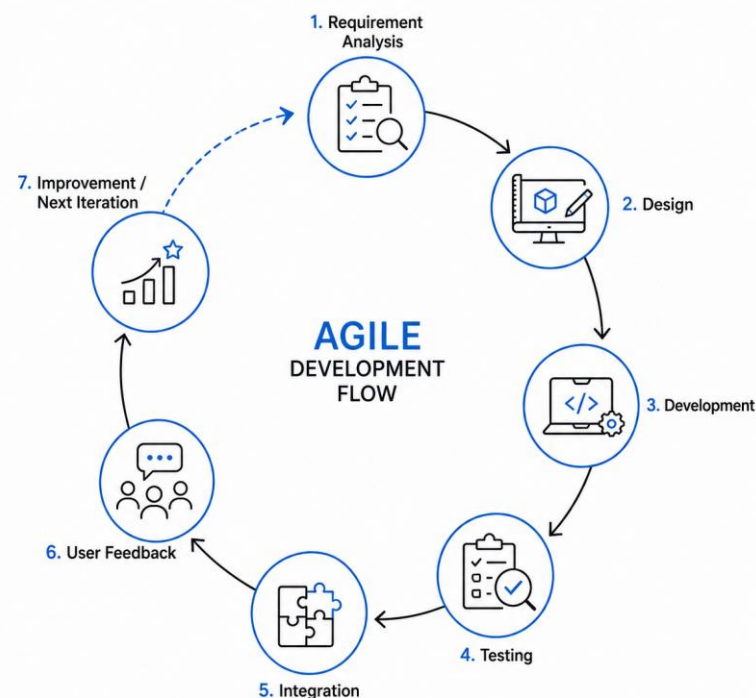
- **Limited Integration:** Existing systems mainly focus on air-quality or individual environmental parameters, rather than combining environmental and safety monitoring.
- **Lack of Visual Detection:** Most systems do not integrate **camera-based AI** for detecting fire, smoke, waterlogging and road damage.
- **Limited Predictive Analysis:** Existing solutions provide monitoring but have limited **AI/ML-based risk prediction**.
- **Weak Incident Reporting:** GPS-based reporting with verified visual evidence is generally limited.
- **No User Verification:** Few systems provide a verification step before generating incident reports.
- **Fragmented Monitoring:** Environmental sensors, visual monitoring and safety alerts are often separate rather than integrated into one platform.

Research Gap: There is a need for an **integrated IoT + AI system** that can *Sense → Detect → Analyze → Predict → Alert → Respond* in real time.

System Comparison

Parameter	PurpleAir	ThingSpeak	Standalone Environmental System	SmartSurround
Primary Purpose	Air-quality monitoring	IoT data analytics	Environmental monitoring	Environmental + safety monitoring
Air Quality	PM monitoring	Depends on sensors	Selected parameters	PM1.0, PM2.5, PM10, VOC, CO ₂
Camera Monitoring	✗	✗	Usually absent	✓ ESP32-CAM
Hazard Detection	✗	✗	Limited	Fire/Smoke, Waterlogging, Road Damage
GPS	Sensor location	Location/channel support	Varies	Incident-based GPS
Real-Time Dashboard	✓	✓	Varies	✓ Integrated dashboard
Historical Data	✓	✓	Depends	✓ Trends & analysis
Predictive Analysis	Limited	Can be developed	Limited	AI/ML-based prediction
Alert System	Air-quality information	Configurable alerts	Threshold-based	Normal → Warning → Critical
Incident Reporting	Limited	✗	Limited	GPS + verified evidence
User Verification	✗	✗	✗	✓ Before reporting
Overall Approach	Air monitoring	IoT analytics	Parameter monitoring	Sense → Detect → Analyze → Predict → Alert → Respond

SDLC Model & Agile Implementation



For SmartSurround, the **Agile SDLC Model** is the most applicable.
Why Agile is Applicable?

Reason	Application in SmartSurround
Incremental Development	Hardware, sensors, AI, backend, and dashboard can be developed in separate iterations.
Continuous Testing	Sensors, ESP32 communication, camera, GPS, and AI detection can be tested individually.
AI Improvement	AI models for fire, smoke, waterlogging, and road damage can be improved through repeated testing.
Easy Integration	Hardware, IoT, AI, database, and web dashboard can be integrated step-by-step.
Flexible Changes	Sensor selection, alert thresholds, dashboard features, and AI requirements can be modified during development.
User Feedback	The dashboard and hazard-reporting workflow can be improved based on testing and user feedback.

Why not Waterfall?

Waterfall is less suitable because SmartSurround involves **AI detection, hardware integration, and iterative testing**. Requirements and performance may need changes after testing the prototype.

Conclusion:

Agile is applicable because SmartSurround requires iterative development, continuous testing, AI improvement, and flexible integration of hardware and software.

Expected Outcomes, Future Scope & Conclusion

Expected Outcomes

- Continuous real-time environmental monitoring
- AI-assisted visual anomaly detection
- GPS-based incident reporting with evidence
- Early warning and predictive analysis
- Role-based access and daily safety reports

Future Scope

- Advanced AI models and edge AI deployment
- Multi-post control center and mobile app
- Smart-city integration and cloud deployment
- Digital map of all posts and incidents

"SmartSurround transforms environmental monitoring from simple data collection into intelligent sensing, detection, prediction, and safety management."

Thank You

Questions & Discussion